

Skill-Adjusted Expected Goals in the NHL: A Two-Phase Machine Learning Framework

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Abstract

Expected goals (xG) models estimate the probability that a shot results in a goal from features such as shot location, angle, and game context. This research project presents a two-phase machine learning pipeline built on approximately 2 million NHL shots (2007–2025). Phase 1’s XGBoost model reached an AUC-ROC of 0.7900, the highest among four classifiers trained on 93 shot-level features. Comparing class-weighted and unweighted configurations revealed that cost-sensitive training is unnecessary at this level of imbalance and actively harms calibration. Removing class weighting reduced log loss by 62% (0.2091 vs. 0.5551) while preserving discrimination. Phase 2 adds shooter finishing skill via Beta-Binomial shrinkage and goalie quality via Goals Saved Above Expected. Yet, these talent features provide marginal gains, with the rolling goalie-only variant coming closest to Phase 1 (AUC: 0.7892 vs. 0.7900). Shot geometry dominates all feature importance analyses, supporting the long-standing recognition of the home-plate danger zone in hockey analytics.

1 Introduction

Historically, hockey performance evaluation has been measured by counting goals and assists. These numbers are straightforward, but they record outcomes rather than opportunities. Expected goals (xG) models address this limitation directly. An xG model takes in the shot’s location, angle, type, and surrounding game context and returns a probability of scoring. When combined over a game or a season, these probabilities provide a measure of the quality of chances created and allowed, independent of random variation in finishing. First introduced in soccer analytics in Barnett and Hilditch’s paper (1993), the concept was later extended to hockey analytics by Macdonald (2012), who proposed expected-goal estimates as replacements for raw goals in adjusted plus-minus calculations. Online prediction models have been rising since then, such as MoneyPuck (Tanner, 2025), Evolving Hockey (Younggren and Younggren, 2018), and HockeyStats.com (JFreshHockey and Bacon, 2025), which will be discussed further in the **Related Work** section.

A recurring debate in xG architecture design is whether to include player identity. Most models purposefully exclude it, since xG should, in theory, measure the opportunity itself rather than who happens to be shooting or tending net. However, in reality, some shooters consistently outperform their expected totals across multiple seasons, and some goalies routinely stop shots that the average goaltender would not

(e.g., Carey Price). If these tendencies are stable, incorporating estimates of shooter finishing and goalie quality into an xG model should improve prediction accuracy. This theory was tested by Noel (2025), who reported improvements of up to 5% in log loss, Brier score, and AUC when separating overall, locational, and situational skill features for shooters and goaltenders relative to a talent-agnostic baseline.

The present project aimed to answer the following research questions:

1. Which machine learning classifier best predicts goal probability from shot-level features?
2. Which features carry the most predictive weight, and do they correspond with what we know about how goals are scored?
3. Does incorporating player-specific talent estimates, such as shooter finishing skill or goalie save performance, considerably improve xG predictions, and what is the best methodology to estimate and include these talent features without introducing bias?

2 Related Work

Macdonald (2012) built an xG model to evaluate National Hockey League (NHL) teams and players using team-level predictors, including faceoffs, hits, goals, shots, missed shots, and blocked shots. He reported a lower mean squared error between actual and predicted goals than preceding approaches. He proposed replacing raw goals with expected-goal estimates in adjusted plus-minus models¹. On the other hand, Found (2016) provided empirical support for goal-based evaluation more broadly: team goal differential predicted winning percentage during the 2015–16 NHL season while shot differential did not, and relative plus-minus per minute of ice time outperformed relative Corsi² per minute as an individual-level metric. These findings suggest that raw shot data are insufficient and thus motivate the use of shot-quality-weighted probability estimates as an evaluation unit.

Looking at public implementations, MoneyPuck (Tanner, 2025) used gradient boosting that trained on over 50,000 goals and 800,000 shots from the 2007–08 through 2014–15 regular seasons and playoffs. They utilized 15 input variables, including shot distance, shot angle, shot type, time since the last event, speed from the previous event, skater counts, power-play state, and an empty-net indicator. Evolving Hockey (Younggren and Younggren, 2018) followed a similar setup, using XGBoost with four strength-state-specific models (even strength, power play, shorthanded, and empty net) and 15 feature groups spanning shot geometry, game state, shot type, and prior-event type³. HockeyStats.com (JFreshHockey and Bacon, 2025) is a newer XGBoost-based public model maintained by JFreshHockey and Patrick Bacon. It was trained on recent seasons (2023–24 through the first half of 2025–26), and used a nested cross-validation procedure with a strict in-season calibration-error constraint of 3%⁴. Johnson (2024) compared four public models (Puckalytics, MoneyPuck, Natural Stat Trick, and Evolving Hockey) on league-wide percent deviation from actual goals

¹Adjusted plus-minus models generally involve the isolation of a player’s individual impact on goal scoring by statistically accounting for the quality of their teammates, opponents, and on-ice situations

²Shots plus missed shots plus blocked shots.

³Cross-validated even-strength AUC is 0.7822, with a 2017–18 holdout AUC of 0.7747.

⁴HockeyStats.com reported an AUC of 0.796 across all situations and 0.800 at even strength.

at 5v5, 5v4, and 4v5 strength states and found that no single model dominated across all situations. These served as the performance benchmarks for the present study.

Several recent academic works have further advanced the approach. For instance, Shmakov (2024) proposed an Expected Goals on Target (XGoT) model using Kontinental Hockey League (KHL) data, which extends the classical xG framework by incorporating the location of each on-target shot on the gate plane itself. It created an explainable complement to the standard goal-probability estimate. Gradient-boosted trees are a practical default for tabular prediction problems of this sort. Chen and Guestrin (2016) introduced XGBoost as a scalable tree-boosting system with regularization and built-in support for class imbalance through the `scale_pos_weight` parameter. Korkee (2025) helped paint a fuller picture by comparing several machine learning algorithms for NHL player performance prediction and found that XGBoost was the most suitable overall and the least prone to overfitting. In contrast, the light gradient-boosted model (LightGBM) produced stronger individual models in some configurations but was more sensitive to overfitting. He also reported that the optimal number of features tended to be around 15 or 30. Town (2025) combined a game-level expected goals model (built with a GAM Poisson specification on predictors including shots on goal, penalty minutes, power play opportunities, faceoffs, and offensive zone-start percentage) with a playoff-success analysis comparing regular-season expected-goal differential against actual goal differential for predicting playoff series outcomes ⁵. Paerels (2020) compared logistic regression, gradient boosting, and generalized additive models (GAM) on nine shot-level predictors (distance, angle, shot type, player differential, overtime, rebound, rush, and inter-conference or inter-division rivalry), and concluded that a GAM with cubic distance and cubic angle terms produced the best AUC. Notably, his findings indicated that the expected goal percentage in overtime was roughly twice that in regular time, i.e., the final 5 minutes.

Noel (2025)’s LightGBM put shooter and goaltender skill into three dimensions: (1) an overall component, (2) a locational component using a shot-binning technique originally proposed by Shuckers and Curro (2013), and (3) a situational component engineered using Gower distance⁶. This method resulted in improvements of up to 5% in log loss, Brier score, and AUC relative to a skill-agnostic baseline. His work inspired the present work to test the same general idea on a larger dataset, to add an ablation to separate shooter and goalie contributions, and investigate the circularity problem and what happens when it could be corrected using a temporal holdout.

Machine learning has been applied to hockey prediction for over a decade. Earlier work, such as Weissbock et al. (2013), trained classification models on traditional and advanced in-game statistics, including features for possession and luck, and found that neural networks produced the most robust single-game predictions with 59.38% accuracy. In that analysis, game location, goals against, and goal differential were the most informative features. In contrast, the PDO statistic⁷ was less informative. The author Weissbock (2014) later extended the work to both single games in regular season and playoff series, and argued that single-game NHL prediction may have a theoretical ceiling of roughly 62% accuracy. Their single-game

⁵NHL regular season consists of 82 games per team, which are used to determine standings. The top teams will then compete in a best-of-seven playoff bracket to determine the Stanley Cup champion.

⁶A measurement that considers the similarity (or dissimilarity) between two sets of variables, accounting for different data types, such as numerical and categorical values.

⁷PDO is the sum of a team’s shooting percentage and save percentage, used to measure how much “luck” or “regression” is influencing the game results.

model reached 59.8% using traditional statistics, while playoff prediction accuracy rose to about 75% with over 30 traditional and advanced features. Sharma (2018) tested whether pair-wise game-level data from the prior season contributes anything to season-level aggregates when predicting a team’s winning percentage, and concluded that the more granular data did not reliably help. Only a limited number of interaction terms involving high goal differential reached significance.

Isolating individual contributions in a team sport is genuinely challenging. Thomas et al. (2013) modeled player ability as a competing stochastic process with separate offensive and defensive parameters for skaters and a defensive-only parameter for goaltenders, applying Laplace-Gaussian pooling⁸ to each position group to shrink individual estimates toward their position mean. Gramacy et al. (2013) argued that traditional plus-minus captures only the marginal effect of individual players and can be noisy because it does not account for sample size. Using regularized logistic regression to estimate each player’s contribution to goal probability, they found that most players did not stand out as measurably strong positive or negative contributors after prior shrinkage. They later formalized this as a regularized logistic regression framework that conditions on on-ice players, teams, and the game situation (Gramacy et al., 2017). Overall, this body of research formed the conceptual foundation for treating player quality as a latent variable that needs appropriate shrinkage.

Bayesian methods have also been applied directly to hockey scoring: Sadeghkhanian and Ahmed (2019) compared two Bayesian models for predicting goal counts in a hockey game, using prior-game points, home advantage, and specialists’ opinions. They later (Sadeghkhanian and Ahmed, 2021) developed a Bayesian predictive density estimator for the waiting time until the r -th goal, modeled with a gamma distribution, with ancillary information via a weighted beta-prime prior, and reported better predictive performance than a minimum-risk-equivariant estimator.

3 Materials and Data Sources

This project used the MoneyPuck Shot Dataset (Tanner, 2025), which covers all NHL shot attempts from the 2007–08 through 2025–26 seasons (including shots as of March 20, 2026). The dataset contains roughly 2 million shot entries, with each record including 124 raw features describing shot geometry, game situation, shot type, and pre-shot context. The target variable is binary: `goal` = 1 if the shot resulted in a goal, and 0 otherwise. Since approximately 8% of shots are goals, the project experimented with both class-weighted and unweighted training to determine whether cost-sensitive learning is needed under this level of imbalance (see Section 5.4).

3.1 Feature engineering

After removing columns with leakage risk, 93 features remain as model inputs. These fall into the following categories:

- *Shot geometry*: Arena-adjusted x and y coordinates, shot distance, shot angle adjusted for shooting side, and arena-adjusted distance and lateral coordinate

⁸Also known as Laplacian of Gaussian (LoG), it is a multi-scale feature extraction technique that combines smoothing and edge detection within a neural network architecture. Commonly used in the context of computer vision and deep learning.

- *Shot type and context*: One-hot encoded shot type (wrist, slap, snap, backhand, tip-in, deflection, wrap-around), binary indicators for rebound and rush situations
- *Game state*: Period, game time, home/away indicator, score differential
- *Skater composition*: Skaters on ice for each team, penalty lengths for each team
- *Pre-shot event context*: Last event type, last event coordinates, time since last event, distance from last event, speed from last event
- *Interaction features*: Computed products of angle and rebound/speed context (e.g., `shotAnglePlusRebound`, `shotAnglePlusReboundSpeed`)

The following column categories were dropped to prevent leakage. First, MoneyPuck’s own xG outputs (`xGoal`, `xFroze`, `xRebound`, and related predictions) were removed, since keeping them would reduce the task to predicting a precomputed score rather than learning from raw features. Second, post-shot outcomes (`shotWasOnGoal`, `shotGoalieFroze`, `shotGeneratedRebound`, `timeUntilNextEvent`) were also dropped as they are only observable after the shot resolves. Third, all identifier and name columns were excluded: `shotID`, team codes, and player name fields. The event sequence number (`id`) and NHL game identifier (`game_id`) needed special handling. They were retained in the data pipeline for sequence construction, but excluded from the model’s feature matrix. A monotonically increasing game identifier implicitly represents temporality, and leaving it in would allow models to learn temporal trends rather than shot-physics patterns. Player identity columns (`shooterPlayerId`, `goalieIdForShot`) were stored separately for Phase 2 talent estimation and not fed as inputs in Phase 1. Note that even though Korkee (2025) recommended 15 or 30 features as optimal feature counts, the full set was retained, as the broader feature matrix enables the feature importance analysis in Section 5.1.

3.2 Temporal split

A strict temporal split of the dataset was used, rather than a random cross-validation. By nature, hockey changes significantly over time: rule modifications, tactical evolution, and league expansion all shift shot distributions in ways that random splits would mask. Training on future seasons and testing on past ones would make the shot prediction too effortless.

The split maintains the following temporal ordering:

- **Training**: Seasons 2007–2020 (~1.47 million shots)
- **Validation**: The 2021 season (~121,000 shots), used for early stopping and configuration selection
- **Test**: Seasons 2022–2025 (~458,000 shots)

The data from the 2021 season was used as validation rather than training because it was played under COVID-19 protocols, with a compressed 56-game schedule and a hub-city format (National Hockey League Players’ Association, 2020), resulting in an atypical distribution. Using it on the validation set reduces the

risk of overfitting to ordinary season patterns, while early stopping on unusual data is acknowledged as a limitation. The test set covers three full post-training seasons, plus the 2025–26 season through March 20, 2026, around the start of this project.

4 Methods

4.1 Phase 1: Baseline models

Four classifiers were trained on the 93 shot-level features utilizing the complete training split, consisting of around 1.47 million shots from the 2007–2020 seasons:

1. **Logistic Regression** was used as a linear baseline model with interpretable coefficients. It was executed through scikit-learn. The strength of L2 regularization was chosen through cross-validation with $C \in \{0.01, 0.1, 1.0, 10.0\}$.
2. **Decision Tree** provided a non-linear yet interpretable baseline. The maximum depth was chosen from $\{3, 5, 10, \text{None}\}$ via cross-validation. Each baseline utilized scikit-learn’s `GridSearchCV` with 3-fold cross-validation and AUC-ROC evaluation.
3. **XGBoost** (Chen and Guestrin, 2016) is an ensemble that utilizes regularized gradient boosting. The number of estimators $\{100, 200\}$, the maximum depth $\{3, 6\}$, and the learning rate $\{0.05, 0.1\}$ were optimized using grid search.
4. **Multi-Layer Perceptron**: Used PyTorch featuring three hidden layers with 128 units per layer, ReLU activation functions, batch normalization following every layer, and a dropout rate of 0.1. Binary cross-entropy was used as the loss function, and the model was optimized with Adam at a learning rate of 0.001 and a batch size of 512. Training was conducted for up to 100 epochs with early stopping (patience set to 10) based on validation loss. A grid search was performed on hidden dimensions $\{64, 128, 256\}$, layer counts $\{2, 3\}$, and dropout rates $\{0.1, 0.3, 0.5\}$ using the validation set. The 3×128 configuration with a dropout of 0.1 yielded the highest validation AUC, whereas deeper or broader architectures did not improve performance.

All experiments used a shared random seed of 42 across Python, NumPy, and PyTorch for reproducibility. As mentioned earlier, to address the 8% class imbalance and to experiment with Chen and Guestrin (2016)’s cost-sensitive training rather than over- or under-sampling, all four classifiers were trained under two configurations: with and without class-weight corrections. In the weighted configuration, logistic regression and the decision tree used scikit-learn’s `class_weight="balanced"`, XGBoost used `scale_pos_weight = nneg/npos  $\approx 12.5$ , and the MLP used an equivalent pos_weight in its binary cross-entropy loss. In the unweighted configuration, all models were trained with default (uniform) class weights. Both configurations are compared in Section 5.1 and calibrated in Section 5.4.`

AUC-ROC⁹ served as the main metric for evaluating results. It assesses the likelihood that a randomly

⁹AUC-ROC is the standard performance metric for classification models. It measures the model’s ability to distinguish between classes across all possible decision thresholds.

selected goal has a higher predicted probability than a randomly selected non-goal. Log loss measures calibration quality by penalizing predictions based on how far the assigned probability is from the true label, rather than just on whether the ranking is correct. Precision, recall, and F1 at threshold 0.5 are reported for reference. SHapley Additive exPlanations (SHAP) values (Lundberg and Lee, 2017) were computed for XGBoost using the TreeExplainer algorithm¹⁰. Unlike tree-impurity-based importance measures, those values are comparable across models and scale with the model output; thus, they can be compared directly across features and individual predictions.

4.2 Phase 2: Talent adjustment

4.2.1 Shooter finishing skill via empirical Bayes

The raw data in per-player shooting could be unreliable due to the inherent variance of small sample sizes. Per NHL stats, shot volume per season is highly heterogeneous, driven by a player’s position, seniority, and cumulative time on ice. While elite offensive players may generate high volumes, a regular NHL player may record only 80 to 150 shots per season. To avoid skewing raw percentages away from a player’s true shooting talent, the empirical Bayes framework (Efron and Morris, 1977) is employed to address this by initially estimating a prior distribution from the population of all shooters, then computing a posterior estimate for each player that is automatically pulled toward the population mean. The mechanism induces shrinkage, drawing estimates for low-volume shooters toward the population mean, while pulling high-volume shooters’ estimates closer to their observed data.

Each player’s goal-scoring is modeled as a Binomial process with shooter-specific success probability θ_i , drawn from a Beta prior (Gelman et al., 2013):

$$\theta_i \sim \text{Beta}(\alpha, \beta) \quad (1)$$

Hyperparameters α and β were calculated from out-of-fold training data by the method of moments. For shooter i with g_i goals on n_i shot attempts, the posterior mean skill estimate is:

$$\hat{\theta}_i = \frac{\alpha + g_i}{\alpha + \beta + n_i} \quad (2)$$

When $\alpha + \beta$ is large relative to n_i , the estimate is pulled toward the prior mean $\alpha/(\alpha + \beta)$. As n_i grows, the individual data takes over. Shooters with fewer than 50 shots would receive the prior mean directly.

4.2.2 Goalie quality via Goals Saved Above Expected (GSAx)

Goals Saved Above Expected (GSAx) for goalie j is:

$$\text{GSAX}_j = \sum_{s \in S_j} \text{goal}_s - \sum_{s \in S_j} \hat{p}_s \quad (3)$$

¹⁰The TreeExplainer algorithm assigns each feature an importance value for a given prediction. It is a class of additive feature-importance measures that have a unique solution and satisfy several desirable theoretical properties.

where S_j is the set of non-empty-net shots encountered by goalie j and $\hat{p}_s$ is the Phase 1 XGBoost’s predicted goal probability for shot s . A negative GSAX means that the goalie is saving more shots than the model expects, given the shot quality. GSAX values were z-score normalized across goalies with at least 50 shots faced. Empty-net shots were assigned a fixed constant of -3.0 to reflect the absence of any goalie.

One important methodological concern in Phase 2 was the source of the talent estimates. The naive approach would be to compute shooter residuals and goalie GSAX from the Phase 1 model’s predictions on the full training set, then proceed to use those values as features in a Phase 2 model trained on the same data, but that would be circular. The talent features encode the Phase 1 model’s training-set error patterns, and the Phase 2 model can relearn those patterns from the talent features without picking up any new signal. In practice, the talent features would be redundant, and Phase 2 would fail to improve on Phase 1. To break this circularity, the training data was split temporally. An inner XGBoost was trained on shots from the 2007–2013 seasons only, and its predictions on the 2014–2020 seasons were used to compute out-of-fold shooter skill and goalie GSAX estimations. Since the inner model never saw the 2014–2020 shots during its own training, the estimates would be independent of Phase 2’s training data. The Phase 2 XGBoost was then trained on the full 2007–2020 data, supplemented with these talent features.

5 Results

5.1 Phase 1 model comparison

Table 1 breaks down the test set performance for all Phase 1 models trained without class weighting on the 2022–2025 test period. XGBoost led with an AUC-ROC of 0.7900 and the lowest log loss at 0.2091.

Model	AUC-ROC	Log Loss	Precision	Recall	F1
Logistic Regression	0.7491	0.2278	0.626	0.067	0.120
Decision Tree	0.7680	0.2438	0.706	0.134	0.225
XGBoost	0.7900	0.2091	0.773	0.150	0.251
MLP (3×128)	0.7845	0.2104	0.789	0.144	0.244

Table 1: Phase 1 model performance on the 2022–2025 test set (without class weighting).

Table 2 compares the class-weighted and unweighted configurations. Removing class weighting either preserved or improved AUC for all models while reducing log loss by 60–65%. The largest AUC gain was with logistic regression (+0.030), while XGBoost improved from 0.7891 to 0.7900. The log-loss reductions were more significant with XGBoost, dropping from 0.5551 to 0.2091, which is a 62% improvement.

Model	With class weighting		Without class weighting	
	AUC	Log Loss	AUC	Log Loss
Logistic Regression	0.7191	0.6435	0.7491	0.2278
Decision Tree	0.7686	0.5872	0.7680	0.2438
XGBoost	0.7891	0.5551	0.7900	0.2091
MLP (3×128)	0.7836	0.5789	0.7845	0.2104

Table 2: Effect of class weighting on Phase 1 models.

Logistic regression performed 4.1% below XGBoost (0.7491 vs. 0.7900). This gap reflects the nonlinear structure of the scoring zone, which a linear model cannot capture. This result matches broader findings in the literature, e.g., Korkee (2025) found that XGBoost was the strongest overall algorithm for hockey prediction.

The ROC curves for all Phase 1 models are shown in Figure 1.

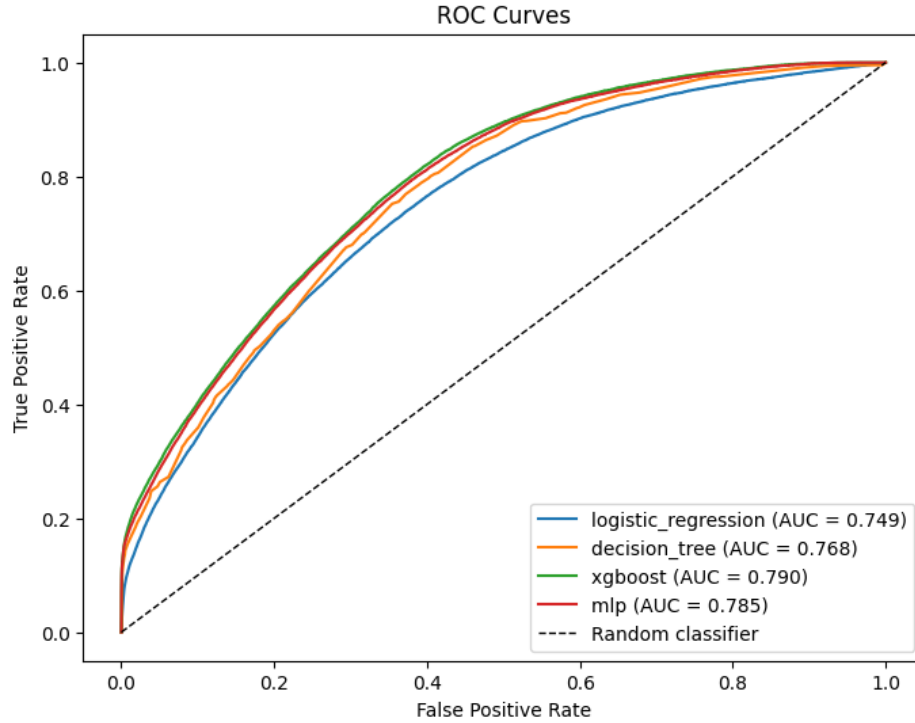


Figure 1: ROC curves for all Phase 1 models on the 2022–2025 test set. AUC values are shown in the legend.

5.2 Feature importance analysis

Table 3 reports SHAP feature importance from XGBoost on the Phase 1 test set. Arena-adjusted shot distance is the most important feature by a wide margin, with roughly twice the mean absolute SHAP value of the next-ranked feature. Three of the top seven features relate to shot geometry. This is consistent with the previous findings in feature rankings, where MoneyPuck (Tanner, 2025) listed shot distance from the net as its top predictor, and Younggren and Younggren (2018) noted that shot distance is the first split in their

decision tree most of the time.

Rank	Feature	Mean SHAP	Group
1	arenaAdjustedShotDistance	0.388	Geometry
2	timeSinceLastEvent	0.221	Pre-shot context
3	homePenalty1TimeLeft	0.210	Penalty state
4	shotDistance	0.197	Geometry
5	distance_x_angle	0.174	Interaction
6	defendingTeamForwardsOnIce	0.160	Skater count
7	arenaAdjustedYCoordAbs	0.152	Geometry
8	homeSkatersOnIce	0.142	Game state
9	homePenalty1Length	0.077	Penalty state
10	shotOnEmptyNet	0.076	Game state

Table 3: Top 10 features by mean absolute SHAP value (XGBoost Phase 1 test set, 93-feature clean model excluding identifier columns)

These rankings match what hockey analysts mentioned when talking about scoring. The low slot area directly in front of the net is where dangerous chances originate. Shot distance from the net (`arenaAdjustedShotDistance`, `shotDistance`) and lateral position (`arenaAdjustedYCoordAbs`) together define the most predictive shooting locations. The model has learned this spatial structure entirely from the data. The penalty features (`homePenalty1TimeLeft`, `homePenalty1Length`) at ranks 3 and 9 capture the higher possible expected goals that come when the home team is at a disadvantage. `timeSinceLastEvent` at rank 2 picks up rebound and rush situations, like when a shot follows quickly after a preceding event, the goalie could still be out of position. `defendingTeamForwardsOnIce` reflects line composition, where fewer defending forwards on ice may indicate weaker defensive coverage. The engineered interaction feature `distance_x_angle` ranks 5th (mean $|\text{SHAP}| = 0.174$), the highest-ranked of the three constructed interaction features. This indicates that the product of shot distance and angle carries a signal beyond either feature alone.

Figures 2 and 3 provide two views of the same SHAP analysis. The beeswarm plot shows the direction and distribution of each feature’s effect across individual shots, whereas the bar chart shows aggregate mean absolute importance.

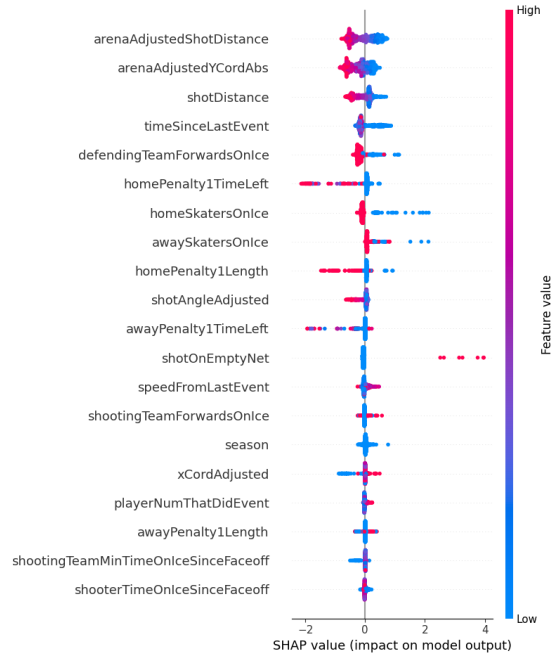


Figure 2: SHAP beeswarm plot for XGBoost (Phase 1). Each point is one test shot. Feature values are color-coded (red = high, blue = low).

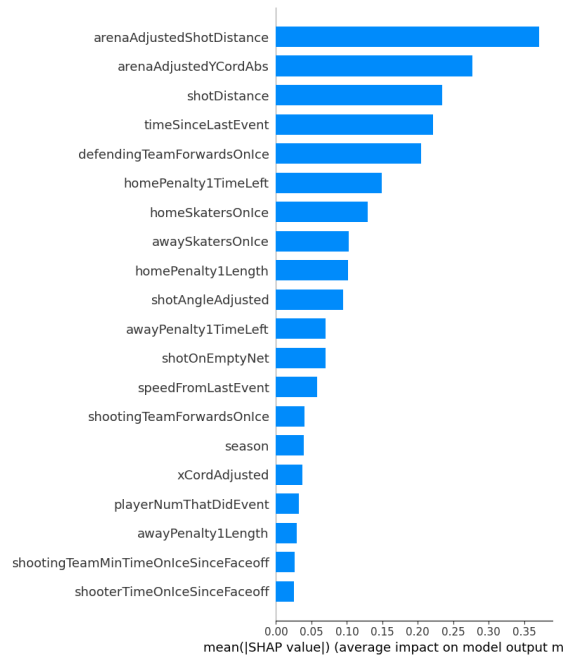


Figure 3: Mean absolute SHAP values for XGBoost (Phase 1), showing the top 20 features by aggregate importance.

5.3 Phase 2 results and ablation

Table 4 presents Phase 2 configurations alongside the Phase 1 XGBoost baseline. The two methods were implemented, including a static pool drawn from the 2014–2020 out-of-fold window and a rolling pool that shifts the estimation window forward to 2018–2021. Within each pool, there is a breakdown of the contributions of shooter skill and goalie quality.

Configuration	AUC-ROC	Log Loss	Δ AUC
Phase 1 XGBoost (no talent)	0.7900	0.2091	—
<i>Static (pool: 2014–2020; $\hat{\alpha} = 8.486$, $\hat{\beta} = 119.039$)</i>			
Shooter skill only	0.7890	—	−0.0010
Goalie quality only	0.7887	—	−0.0013
Both	0.7890	0.2096	−0.0010
<i>Rolling (pool: 2018–2021; $\hat{\alpha} = 4.963$, $\hat{\beta} = 65.407$)</i>			
Shooter skill only	0.7883	—	−0.0017
Goalie quality only	0.7892	—	−0.0008
Both	0.7888	0.2098	−0.0012

Table 4: Phase 2 talent-adjustment results on the 2022–2025 test set.

Under the static configuration, Phase 2 showed a small reduction in discrimination (AUC 0.7890 vs. 0.7900, $\Delta = -0.0010$), so technically there was no meaningful calibration benefit (log loss 0.2096 vs. 0.2091). The rolling-window configuration shifted the estimation window forward by training the inner model on seasons ≤ 2017 , with out-of-fold predictions covering 2018–2020, and by adding Phase 1 XGBoost predictions for the 2021 validation season to the pool. It showed mixed effects at the component level: goalie quality improved relative to the static pool ($\Delta = -0.0008$ vs. -0.0013), implying that fresher estimates retain more predictive power. Shooter skill, however, worsened ($\Delta = -0.0017$ vs. -0.0010).

All in all, the Phase 2 configuration did not exceed Phase 1. The rolling goalie-only variant came closest ($\Delta = -0.0008$), but the gap remained. This outcome differed from that of the class-weighted models, in which the rolling goalie-only variant appeared to surpass Phase 1 by a small margin (see Table 2). Once Phase 1 probabilities were adequately calibrated, the residual talent signal was too small to overcome the noise introduced by the temporal gap between estimation and evaluation (See Section 6.1 and Section 5.4 for more detailed discussion).

Figure 4 shows SHAP feature importance for the Phase 2 model, with the two talent features marked in orange. Both `shooter_skill` and `goalie_skill` rank near the bottom of the top-20 list. Shot geometry still dominates. The model comprehended to use the talent features but assigned them relatively little weight, which aligns with the small AUC changes in Table 4.

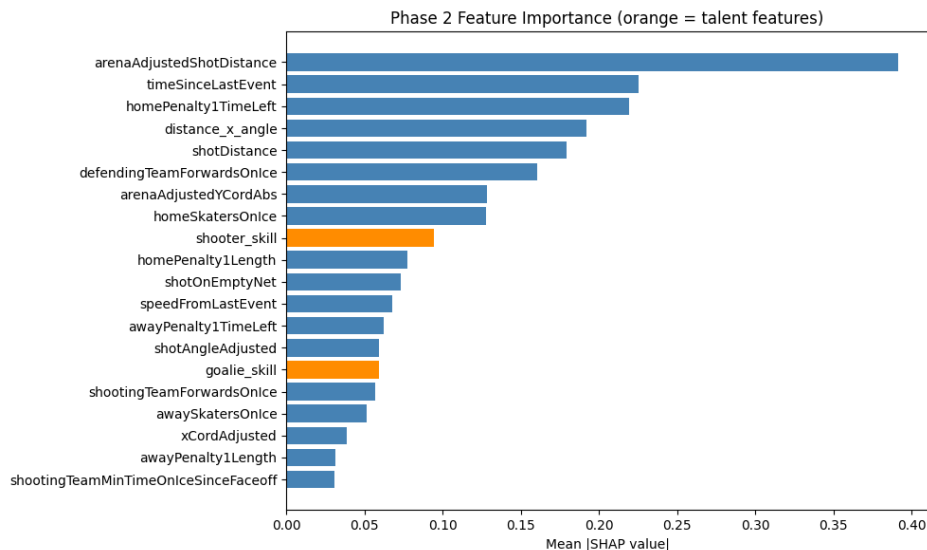


Figure 4: SHAP feature importance for the Phase 2 XGBoost model.

5.4 Calibration

The plot on the right of figure 5 shows calibration curves for the Phase 1 models trained without class weighting, calculated with quantile binning with 15 equal-count bins. The class-weighted models on the left, by contrast, produced calibration curves far from the diagonal. With `scale_pos_weight` ≈ 12.5 , XGBoost’s mean predicted probability was ~ 0.51 rather than the true base rate of 0.07, increasing every probability estimate by about a factor of seven. Removing class weighting could eliminate the distortion, and this result is consistent with the findings of Wallace and Dahabreh (2014), who demonstrated that class-weight adjustments designed for imbalanced data can harm probability calibration without improving ranking performance for tree-based models. This also corresponds with King and Zeng (2001), who finalized an analytic correction for the probability distortion introduced by case-control and prior-correction methods in rare-event settings.

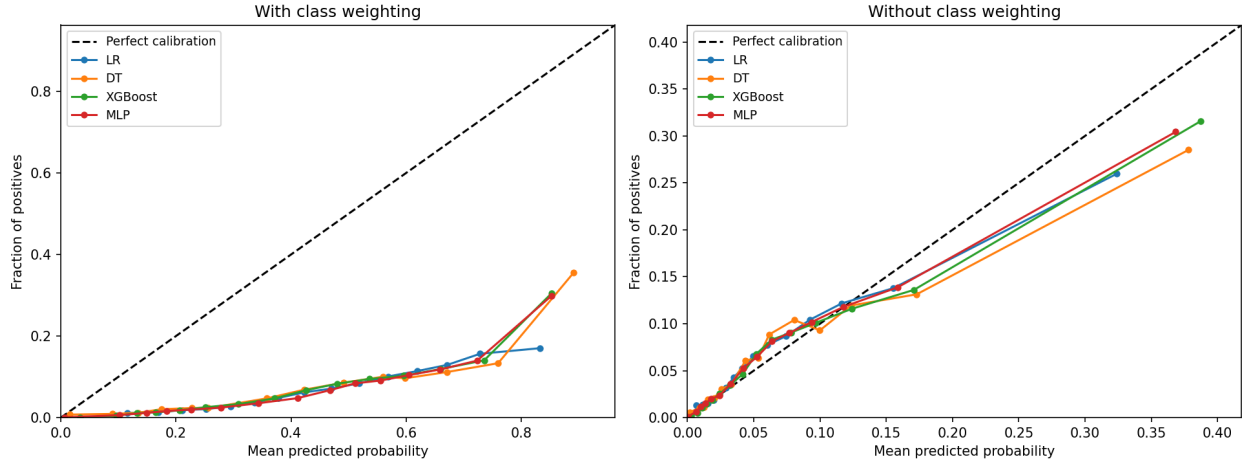


Figure 5: Calibration curves with class weighting (left) vs. without class weighting (right).

6 Discussion

XGBoost was the best performing Phase 1 model, with an AUC of 0.7900, which falls within the same range as publicly reported xG models. For reference, the even-strength XGBoost model of Younggren and Younggren (2018) has a cross-validated AUC of 0.7822 and a 2017–18 holdout AUC of 0.7747, and HockeyStats.com (JFreshHockey and Bacon, 2025) reported 0.796 across all situations and 0.800 at even strength¹¹. In terms of class weighting, all four classifiers exhibited the same pattern after removing that configuration (Table 2). This decision reduced the minority-class loss, shifting the decision boundary to prioritize recall at low thresholds. For AUC, which is integrated over all thresholds, this shift is largely irrelevant. Though the probability estimates, which are the xG values, became systematically inflated. Results also showed that tree-based models with sufficient depth could handle moderate imbalance (7–8% positive rate) without explicit correction, as individual leaves naturally depict low base-rate probabilities.

Shooter finishing skill varied, but the signal it offered required a large sample size. Method-of-moments estimation on the out-of-fold training data (2014–2020 seasons) rendered Beta-Binomial hyperparameters $\hat{\alpha} = 8.486$ and $\hat{\beta} = 119.039$. The prior mean shooting rate was $\hat{\alpha}/(\hat{\alpha} + \hat{\beta}) = 8.486/127.525 \approx 6.7\%$, which is not too far from the empirical 8% goal rate in the full data. There were 2,410 shooters with at least one shot in the out-of-fold period being calibrated, and 278 goalies who faced at least 50 non-empty-net shots and obtained individual GSAX estimates. The practical implication of these parameters is heavy shrinkage, i.e., a shooter needs approximately 128 attempts before their individual data bears equal weight with the prior ($\alpha + \beta \approx 127.5$). With 100 shots, the posterior is about 56% prior and 44% individual data. To sum up, finishing talent is variable enough that small samples mislead more often than they inform, so a heavy shrinkage is reasonable.

¹¹MoneyPuck (Tanner, 2025) does not publicly report AUC, and Johnson (2024) compared public models on league-wide percent deviation from actual goals rather than AUC. Hence, these numbers are more indicative than directly comparable.

6.1 The effect of talent pool recency

The original static Phase 2 configuration calculated talent estimates from 2014–2020 out-of-fold data, so the shots were 2 to 8 years old by the time they were applied to the 2022–2025 test shots. The rolling-window configuration shifted the pool to 2018–2021 instead, with the inner model trained on ≤ 2017 seasons, out-of-fold predictions covered 2018–2020, and Phase 1 XGBoost predictions for the 2021 validation season (which Phase 1 never saw during training). This means that the combined pool is 1 to 4 years old at the time of testing, rather than 2 to 8 years old.

The recency effect is component-specific more than uniform. For goalie quality, the rolling window narrows the AUC gap from -0.0013 (static) to -0.0008 (rolling), consistent with the expectation that fresher estimates track current ability more accurately. However, for shooter skill, the pattern reverses: the rolling window widens the gap from -0.0010 to -0.0017 , suggesting that the shorter pool potentially introduced more noise than signal for shooters. This asymmetry has a plausible explanation that comes down to how hockey and the NHL work. Player careers change substantially over multi-year time frames. Players (shooters) who were at the peak of their careers (or at least, productive) in 2017 may have aged, been injured, or even retired by 2023. A shorter pool also means fewer shots per player, amplifying sampling noise for a quantity (shooting percentage) that is already highly variable. Goalie quality, on the other hand, seems to be more stable from year to year and benefits more directly from recency. Moreover, rookies or younger players appearing during the 2022–2025 test period are assigned the population means regardless of pool vintage, which also added noise in both configurations.

The lesson learned is that when deploying a talent-adjusted xG model, the choice of pool vintage should be guided by which talent component matters most for the application at hand. Goalie-focused adjustments benefit from shorter, more recent windows, whereas shooter-focused adjustments may require longer pools to accumulate enough data per player.

6.2 The circularity problem

The circularity issue has implications beyond this study’s particular application. Whenever a new feature is derived from one model’s residuals and then fed into a second model trained on the same data, there exists a risk that the feature would encode the first model’s training-set error pattern rather than a generalizable signal. On training data, the first model’s residuals are correlated with exactly the cases where the second model requires support. Hence, the feature may appear useful during training, but, unfortunately, may add nothing practical during testing. The fix was to compute derived features from data or model outputs independent of the downstream training set. A temporal holdout provided the independence used here. In cross-sectional settings, a held-out fold split by *observation ID* would have served the same purpose. This problem could be easy to miss in practice: if Phase 2 is trained on the same data as Phase 1, and shows no improvement from a feature that should technically help, the issue may not be that the feature is uninformative. It may be that the estimation method depends on the training data. At the same time, fixing the circularity is necessary but not sufficient on its own. The talent signal must also extend across the temporal gap between training and test. Both conditions need to be controlled for in Phase 2 to outperform Phase 1.

7 Conclusion

The present project found that XGBoost was the best Phase 1 model (AUC 0.7900), outperforming logistic regression (0.7491) and MLP (0.7845), competitive with public xG benchmarks. Shot geometry dominates feature importance, as expected. Cost-sensitive learning could be unnecessary and harm calibration. In this experiment, removing class weighting reduced log loss by 62% while preserving or improving AUC across all classifiers. With properly calibrated Phase 1 probabilities, Phase 2 talent features yield only marginal gains, with no configuration exceeding Phase 1 discrimination (best: AUC 0.7892 vs. 0.7900). These results imply a few generalizable lessons. Talent features derived from training-set residuals suffer from circularity, whereas a temporal holdout is needed to break that dependency. Furthermore, goalie quality transfers more reliably than shooter finishing skill, which could be noisier and thus requires larger samples to estimate.

7.1 Limitations and Future Work

Both Phase 2 configurations treat each player’s skill as designated within the estimation window, ignoring within-window career trajectories. The AUC differences between Phase 1 and Phase 2 configurations are small (0.0008–0.0017) and are reported as point estimates without bootstrap confidence intervals. Given the test set size of ~458,000 shots, the standard error of AUC under bootstrap sampling is typically 10^{-3} , so none of the Phase 2 results can be statistically distinguished from Phase 1 at established significance thresholds. Upcoming work should include bootstrap or permutation-based inference to formalize these comparisons. Also, a player whose performance improved steadily from 2018 to 2021 is represented by a period average rather than a trend. Bayesian state-space models, which allow sequential updating of skill estimates as new data arrive, could be a tool for addressing this limitation.

For future research directions, this work can be extended to incorporate richer spatial information about shots. As Shmakov (2024) proposed, an Expected Goals on Target (XGoT) model, which augments a classical xG estimate with the point on the gate plane at which each on-target shot would cross, could produce an explicable supplement to the standard goal-probability estimate. More broadly, player-tracking data from the NHL’s expanding puck-and-player-tracking infrastructure may eventually provide fine-grained spatial features (defender proximity, goalie positioning) that shot-event logs cannot capture. Such data, using Computer Vision (CV) models, would enable the construction of new predictors for shooting-lane obstruction, relative skater proximity, and instantaneous goalie positioning at the moment of the shot, potentially improving prediction accuracy.

For talent estimation, a dynamic model that tracks player skill progressions over time would be more appropriate than a static lookup table. The rolling-window results in this project suggest that even a simple two- to three-season recency window captures most of the available gain, so a full trajectory model following career arcs should do better in theory. Bayesian methods developed for related hockey problems, including the scoring-number and scoring-time models of Sadeghkhanian and Ahmed (2019, 2021), offer potentially useful building blocks. The goal-importance framework of Lambrix and Carlsson (2022); Lambrix et al. (2025), which weights goals by their contribution to team wins and ties using context variables such as time remaining, manpower differential, and goal differential, offers a relevant extension that instead of treating all

goals as equally valuable, an xG model could weight shots by the leverage of the game situation, producing a metric more directly tied to winning.

References

- Barnett, V. and Hilditch, S. (1993). The effect of an artificial pitch surface on home team performance in football (soccer). *Journal of the Royal Statistical Society. Series A (Statistics in Society)*, 156(1):39–50.
- Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794.
- Efron, B. and Morris, C. (1977). Stein’s paradox in statistics. *Scientific American*, 236(5):119–127.
- Found, R. (2016). Goal-based metrics better than shot-based metrics at predicting hockey success. *The Sport Journal*, 20.
- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis*. Chapman and Hall/CRC, 3rd edition.
- Gramacy, R. B., Jensen, S. T., and Taddy, M. (2013). Estimating player contribution in hockey with regularized logistic regression. *Journal of Quantitative Analysis in Sports*, 9(1):97–111.
- Gramacy, R. B., Taddy, M., and Tian, S. (2017). Hockey player performance via regularized logistic regression. In *Handbook of Statistical Methods and Analyses in Sports*, pages 303–322. Chapman and Hall/CRC.
- JFreshHockey and Bacon, P. (2025). Expected goals model methodology. HockeyStats.com. <https://hockeystats.com/methodology/expected-goals>.
- Johnson, D. (2024). Quick comparison of four public expected goal models. Hockey Analysis. <https://hockeyanalysis.com/2024/04/08/quick-comparison-of-four-public-expected-goal-models/>.
- King, G. and Zeng, L. (2001). Logistic regression in rare events data. *Political Analysis*, 9(2):137–163.
- Korkee, L. (2025). Using machine learning for player performance prediction in ice hockey games. Master’s thesis.
- Lambrix, P. and Carlsson, N. (2022). Performance metrics for ice hockey accounting for goal importance. In *Linköping Hockey Analytics Conference*, pages 11–25.
- Lambrix, P., Carlsson, N., and Säfvenberg, R. (2025). Goal-based performance metrics for ice hockey accounting for goal importance. *Journal of Sports Analytics*, 11:22150218241290987.
- Lundberg, S. M. and Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, pages 4765–4774.

- Macdonald, B. (2012). An expected goals model for evaluating NHL teams and players. In *MIT Sloan Sports Analytics Conference*.
- National Hockey League Players' Association (2020). NHL, NHLPA announce plans for 2020–21 regular season and playoffs. News release. <https://www.nhlpa.com/news/nhl-nhlpa-announce-plans-for-2020-21-regular-season-and-playoffs/>.
- Noel, J. T. P. (2025). Expected by whom? a skill-adjusted expected goals model for NHL shooters and goaltenders. arXiv preprint arXiv:2511.07703.
- Paerels, T. (2020). Play for the point or go for the win: Expected goals in the national hockey league. Master's thesis.
- Sadeghkhani, A. and Ahmed, S. E. (2019). A Bayesian approach to predict the number of goals in hockey. *Stats*, 2(2):228–238.
- Sadeghkhani, A. and Ahmed, S. E. (2021). Predicting the scoring time in hockey. *Journal of Statistical Theory and Practice*, 15(2):43.
- Schuckers, M. and Curro, J. (2013). Total hockey rating (thor): A comprehensive statistical rating of national hockey league forwards and defensemen based upon all on-ice events. In *MIT Sloan Sports Analytics Conference*, Boston, MA.
- Sharma, S. (2018). Predicting the current season's win percentages in the national hockey league using data from the previous season: Can game-level data help? Senior thesis, Claremont McKenna College. CMC Senior Theses, 1990.
- Shmakov, N. (2024). Application and development of the expected goals for hockey player evaluation. In *International Sports Analytics Conference and Exhibition*, pages 235–248. Springer Nature Switzerland.
- Tanner, P. (2025). MoneyPuck expected goals methodology. <https://moneypuck.com/about.htm>.
- Thomas, A. C., Ventura, S. L., Jensen, S. T., and Ma, S. (2013). Competing process hazard function models for player ratings in ice hockey. *The Annals of Applied Statistics*, 7(3):1497–1524.
- Town, A. (2025). Predicting expected goals and playoff success in the national hockey league. Senior thesis, New College of Florida. Theses & ETDs, 6644.
- Wallace, B. C. and Dahabreh, I. J. (2014). Improving class probability estimates for imbalanced data. *Knowledge and Information Systems*, 41(1):33–52.
- Weissbock, J. (2014). *Forecasting Success in the National Hockey League Using In-Game Statistics and Textual Data*. PhD thesis, University of Ottawa.
- Weissbock, J., Viktor, H. L., and Inkpen, D. (2013). Use of performance metrics to forecast success in the national hockey league. In *MLSA@PKDD/ECML*, pages 39–48.

Younggren, J. and Younggren, L. (2018). A new expected goals model for predicting goals in the NHL. Evolving Hockey. <https://evolving-hockey.com/blog/a-new-expected-goals-model-for-predicting-goals-in-the-nhl/>.

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Go Habs Go!

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